Hardware Application Note

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INNOVATOR IN ELECTRONICS

Type 2LL/2KL
Wi-Fi™ + Bluetooth® + 802.15.4(2LL Only) Module

NXP IW610G/IW610F Chipset for
802.11a/b/g/n/ac/ax + Bluetooth 5.4 + IEEE802.15.4 (IW610G only)
Hardware Application Note - Rev. 3.0

- NXP Chipset: IW610G / IW610F
- Design Name: Type 2LL / Type 2KL
- P/N: LBES0ZZ2LL-001 / LBEE0ZZ2KL-001

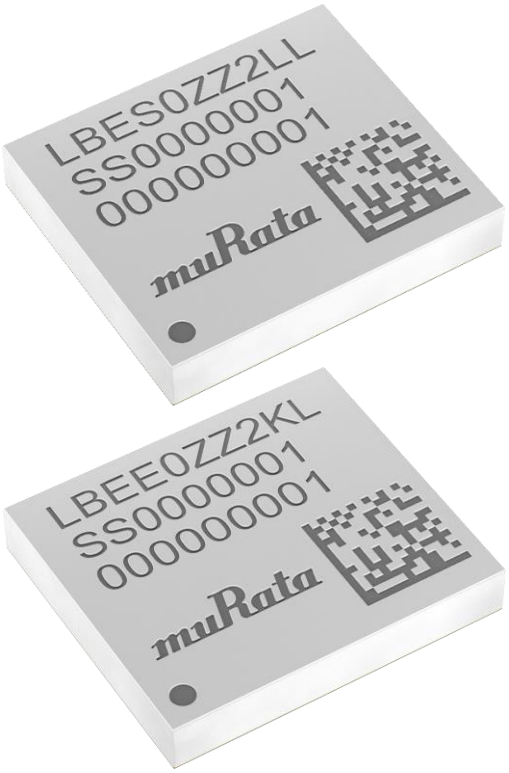


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About This Document

Murata's Type 2LL is a small and very high-performance module based on NXP's IW610G combo chipset, supporting IEEE 802.11a/b/g/n/ac/ax(HE20) + Bluetooth 5.4 LE + 802.15.4.
Murata's Type 2KL is a small and very high-performance module based on NXP's IW610F combo chipset, supporting IEEE 802.11a/b/g/n/ac/ax(HE20) + Bluetooth 5.4 LE.
This application note provides RF and hardware design guidance. Refer to Type 2LL Datasheet and Type 2KL Datasheet for module specification.

Audience & Purpose

Intended audience includes any customer looking to integrate this module into their product. In particular, RF, hardware, systems, and software engineers.

Document Conventions

Table 1 describes the document conventions.

Table 1: Document Conventions

Conventions	Description
	**Warning Note** Indicates very important note. Users are strongly recommended to review.
	**Info Note** Intended for informational purposes. Users should review.
	**Menu Reference** Indicates menu navigation instructions. **Example**: Insert➔Tables➔Quick Tables➔Save Selection to Gallery
	**External Hyperlink** This symbol indicates a hyperlink to an external document or website. **Example**: Murata Click on the text to open the external link.
	**Internal Hyperlink** This symbol indicates a hyperlink within the document. **Example**: Scope Click on the text to open the link.
`Console input/output or code snippet`	**Console I/O or Code Snippet** This text ***Style*** denotes console input/output or a code snippet.
`# Console I/O comment` `// Code snippet comment`	**Console I/O or Code Snippet Comment** This text ***Style*** denotes a console input/output or code snippet comment. • Console I/O comment (preceded by "#") is for informational purposes only and does not denote actual console input/output. • Code Snippet comment (preceded by "//") may exist in the original code.

1 Scope

This application note provides detailed information on schematic/layout design, and references RF performance benchmarks. Refer to Type 2LL Datasheet and Type 2KL Datasheet for module specification.

2 Module Introduction

Type 2LL/2KL is a small and very high-performance module based on NXP IW610G/IW610F combo chipset which supports Wi-Fi 802.11a/b/g/n/ac/ax(HE20) + Bluetooth 5.4 LE + 802.15.4 (2LL only) up to 115 Mbps PHY data rate on Wi-Fi and 2 Mbps PHY data rate on Bluetooth Low Energy.

The WLAN section supports SDIO 3.0 interface, with optional support for USB 2.0 interface. The Bluetooth section supports high-speed 4-wire UART interface, with optional support for USB interface. The 802.15.4 section supports SPI interface.

The IW610G/IW610F implements sophisticated enhanced collaborative coexistence hardware mechanisms and algorithms, which ensure that WLAN and Bluetooth and 802.15.4 (IW610G only) collaboration is optimized for maximum performance.

In IEEE 802.11ax mode, the WLAN operation supports rates of MCS0 – MCS9 in 20 MHz for data rate up to 115 Mbps.

2.1 Features

- WLAN 802.11a/b/g/n/ac/ax 1x1 SISO + Bluetooth Low Energy (Version 5.4) + 802.15.4 combo SMD module with NXP IW610.
- Small size LGA package with resin molding and metal shielding.
- Host interfaces: SDIO 3.0 and USB 2.0 for WLAN; HCI UART and USB 2.0 for Bluetooth; SPI for 802.15.4.
- WLAN MAC address and BD address and 802.15.4 MAC address are stored in OTP

802.15.4 only for 2LL/IW610G

2.2 Hardware Block Diagrams

This section shows the block diagram for Type 2LL / 2KL modules. **Figure 1** shows Type2LL block diagram and Figure 2 shows Type 2KL block diagram.

Figure 1: Block Diagram - Type 2LL

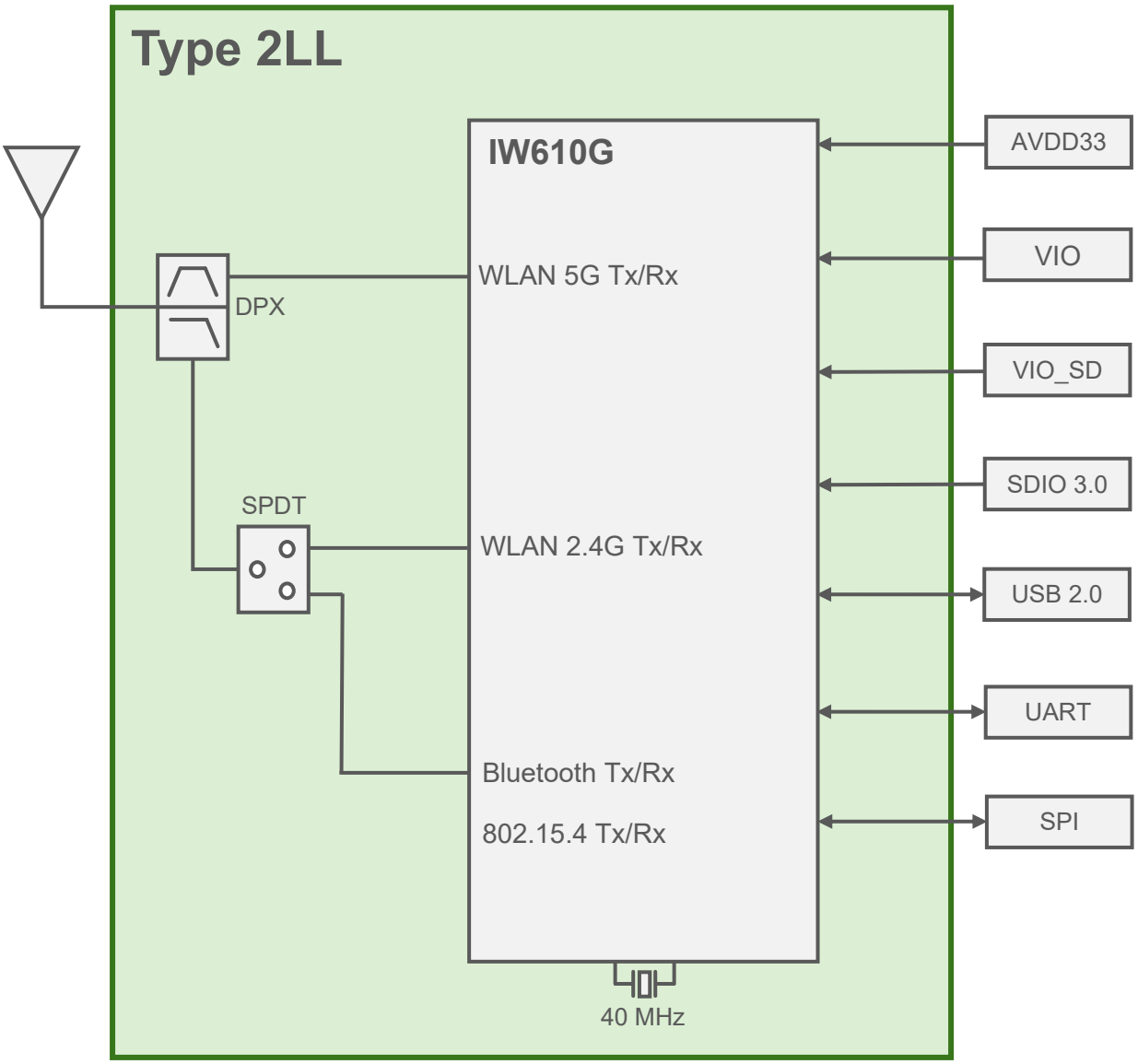
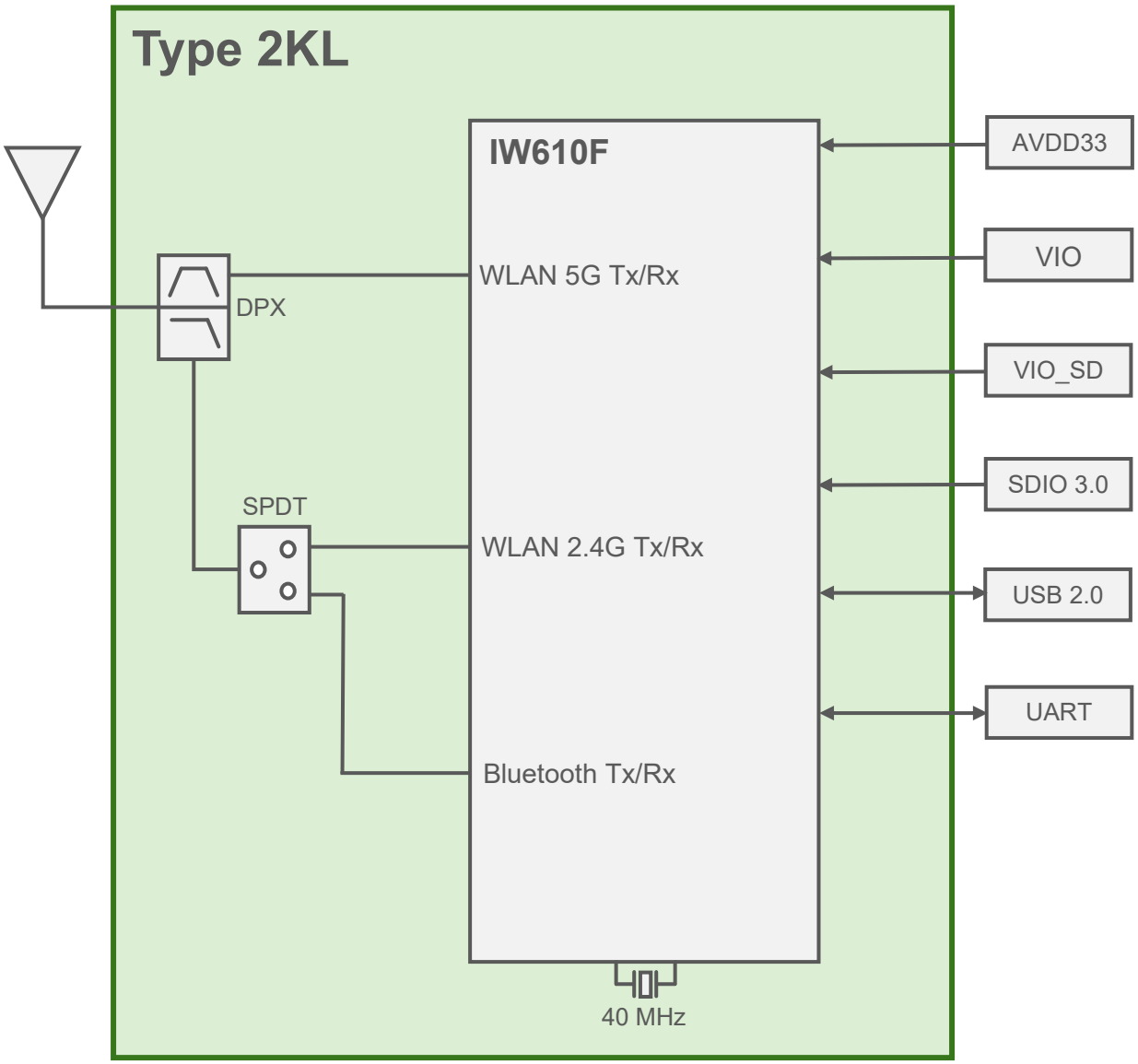


Figure 2: Block Diagram - Type 2KL



3 Reference Design

This section details reference schematics which the end user can leverage for designing their own hardware. Two implementations are shown: one using the PCB trace antenna; the other using u.FL or MHF connectors.

3.1 Reference Circuit

Figure 3 shows the u.FL/MHF connector and the trace antenna for Type 2LL module. **Figure 4** shows the u.FL/MHF connector and the trace antenna for Type 2KL module.

Figure 3: Reference Circuit - Type 2LL

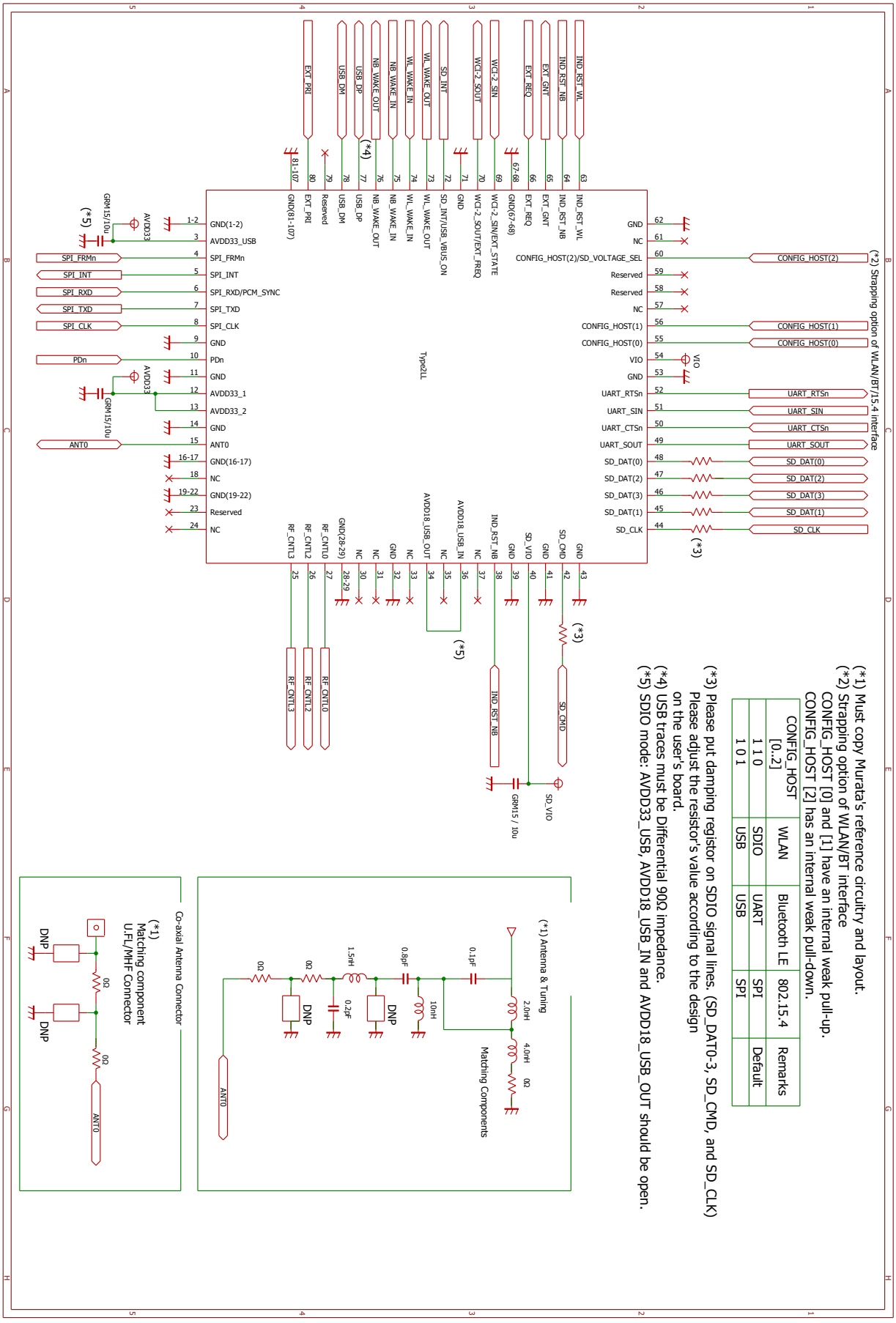
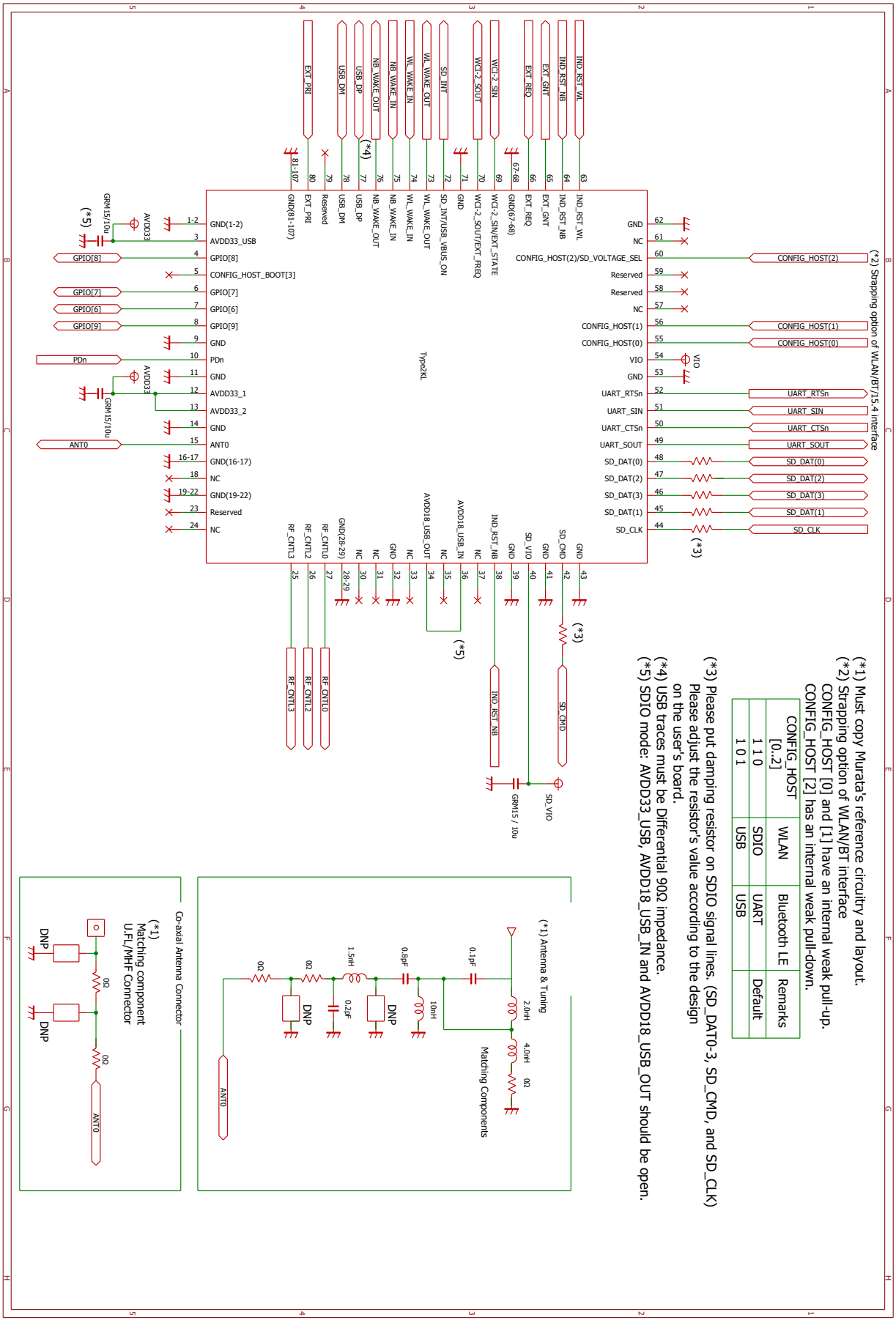


Figure 4: Reference Circuit - Type 2KL



3.2 Requirement for SDIO Signals

SDIO traces should be isometric zero delay routing with 50Ω impedance.

Table 2: SDIO Internal Pull-Up/Pull-Down Specifications

Parameter	Minimum	Typical	Maximum	Unit
Internal nominal pull-up/pull-down resistance	70	100	140	kΩ

3.3 Requirement for USB Signals

USB DP/DM is differential 90 Ω impedance.

3.4 Requirements for Unused Signals

If these signals are not used, no pull-up/down is necessary (floating) for:

- EXT_***
- IND_RST_***
- NB_WAKE_***
- RF_CNTL*
- WCI-2_***
- WL_WAKE_***
- XOSC_EN

In SDIO mode, Floating the unused USB related pins for:

- AVDD33_USB
- AVDD18_USB_OUT
- AVDD18_USB_IN
- USB_DP
- USB_DN

In USB mode, Floating the unused SDIO and UART related pins for:

- SD_CLK
- SD_CMD
- SD_DAT[*]
- SD_VIO
- UART_****

3.5 Module Footprint Design

Refer to dimensions in the Type 2LL Datasheet / Type 2KL Datasheet.The DXF File of module footprint is provided via website.

3.6 Recommended Antenna

This module is certified with two types of antenna solution by regulatory certification body. To use Murata's regulatory certification, any user must follow the instructions below. The DXF File of the trace antenna is provided via website.

3.6.1 PCB Type Di-pole Antenna with the Co-axial Connector

- Users must use recommended antennas are shown in **Table 3**. However, user can use any equivalent type of antenna with less antenna gain than antenna gain of recommended antennas for US/CA under approval of Class I Permissive Change by Murata.
- JP is required to add an antenna in the Certified antenna list if user use an antenna that is not in the certified antenna list in **Table 3** and **Table 4**.
- EU can use the antenna list in **Table 3** and **Table 4**. If the antenna gain is lower than the maximum specified in **Table 4**, it is permissible to select antennas other than those listed.

Table 3: Cable Options for Antenna Gains

P/N	Vendor	Form Factor	Type	Peak Gain [dBi]		Cable Options
				2.4 GHz	5 GHz	
146187	Molex	u.FL/PCB	Dipole	3.2	4.25	50 mm or more
146153	Molex	u.FL/PCB	Dipole	3.2	4.25	50 mm or more
219611	Molex	u.FL/PCB	Dipole	2.67	3.67	40 mm or more
WT32D1-KX	Unictron	MHF I/PCB	Dipole	3.0	4.0	119 mm or more
AA258	Unictron	MHF I/PCB	Dipole	0.9	3.8	100 mm or more
FXP830.07.0100C	Taoglas	MHF I/PCB	Dipole	2.5	4.7	100 mm or more
SRF2W012	Antenova	MHF I/PCB	Dipole	3.0	4.0	100 mm or more
STDANTEMD-013	Sansei	MHF I/PCB	Dipole	1.9	2.8	120 mm
EMF2471A3S	Laird	MHF I/PCB	Dipole	2.4	4.4	100 mm
1001932FT	KAVX	MHF I/PCB	Dipole	2.5	4.4	100 mm or more
1001932PT	KAVX	MHF I/PCB	Dipole	2.03	4.45	100 mm or more
W3P35x8W04	KAVX	MHF I/PCB	Dipole	2.3	5.0	100 mm or more
W2.4-5P-U	Inventek	u.FL/PCB	Dipole	3.73	5.18	90 mm or more

If using an antenna vendor that is not listed (Pulse, TE, etc.) with the Class I Permissive Change, please evaluate the antenna complies with the technical rules of the FCC KDB 178919 Permissive Change Policy.

Table 4: Additional Registered Antennas for Japan and EU

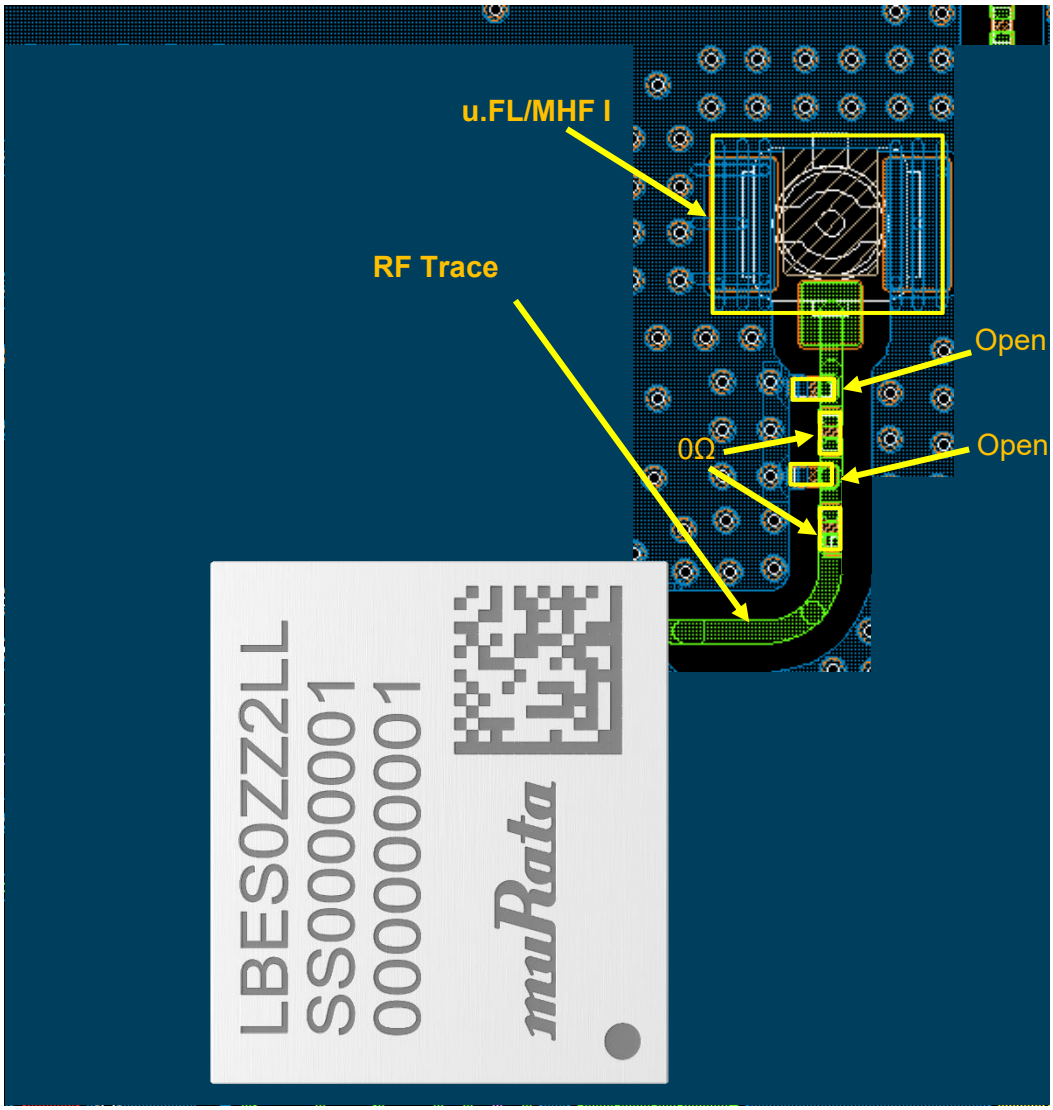
P/N	Vendor	Form factor	Type	Peak Gain [dBi]		Cable Options
				2.4 GHz	5 GHz	
TA-38B-A-W01	Unictron	RSMA	Dipole	2.61	3.07	
TA-6B-A-WE01	Unictron	RSMA	Dipole	2.23	4.59	
GW.48.A151	Taoglas	RSMA	Dipole	1.82	3.28	
GW.40.2153	Taoglas	RSMA	Dipole	3.15	4.39	
GW.52.A153	Taoglas	RSMA	Dipole	2.93	4.53	
W24P-U	Inventek	u.FL/PCB	Dipole	3.58	N/A	
W2.4-5P-U	Inventek	u.FL/PCB	Dipole	3.73	3.79(JP) / 5.18(EU)	90 mm or more

W2.4-5P-U: In Japan, the maximum gain is 3.79 dBi for frequencies up to 5.75 GHz. However, in Europe, FCC, and ISED, the gain is 5.18 dBi for support up to 5.85 GHz.

- Users must copy RF trace to u.FL/MHF I connector from the trace layout file provided by Murata in adherence to the guidelines on:
  - Trace width accuracy within +/- 0.25 mm.
  - Stack height between GND layer and RF trace of 230 ~ 240 µm (Exclude inaccuracy of PCB).
  - Passive component location matching Murata design.
  - Necessary “Keep out” area around u.FL/MHF connector.

Figure 5 show the PCB Type di-pole antenna for Type 2LL/2KL module.

Figure 5: PCB Type Di-pole Antenna - Type 2LL/2KL



Size: 0603 LQP03 / GRM03 / Resistor.

3.6.2 Trace Antenna

- Users must follow the antenna guidelines listed below:
  - Copy antenna design from the antenna layout file provided by Murata.
  - Copy RF trace to PCB trace antenna from the trace layout file provided by Murata adhering to the guidelines listed below:
    - Trace width accuracy within +/- 0.25 mm.
    - PCB thickness within 0.6 ~ 1.6 mm range (0.8 mm typ.).
    - Stack height between GND layer and RF trace of 235 µm; keeping inaccuracy within +/-0.5 µm.
    - Passive component location matching Murata design.

Table 5 shows the trace antenna gain of Type 2LL module.

Table 5: Trace Antenna Gain

Part Number	Vendor	Form factor	Type	Peak Gain [dBi]	
				2.4GHz	5GHz
Type2LL_Antenna	Murata	Trace	Monopole	3.6	4.6

Size: 0603 LQP03 / GRM03 / Resistor

Figure 6 show the trace antenna guidelines for Type 2LL/2KL module.

Figure 6: Trace Antenna Guidelines - Type 2LL/2KL for Shared Antenna

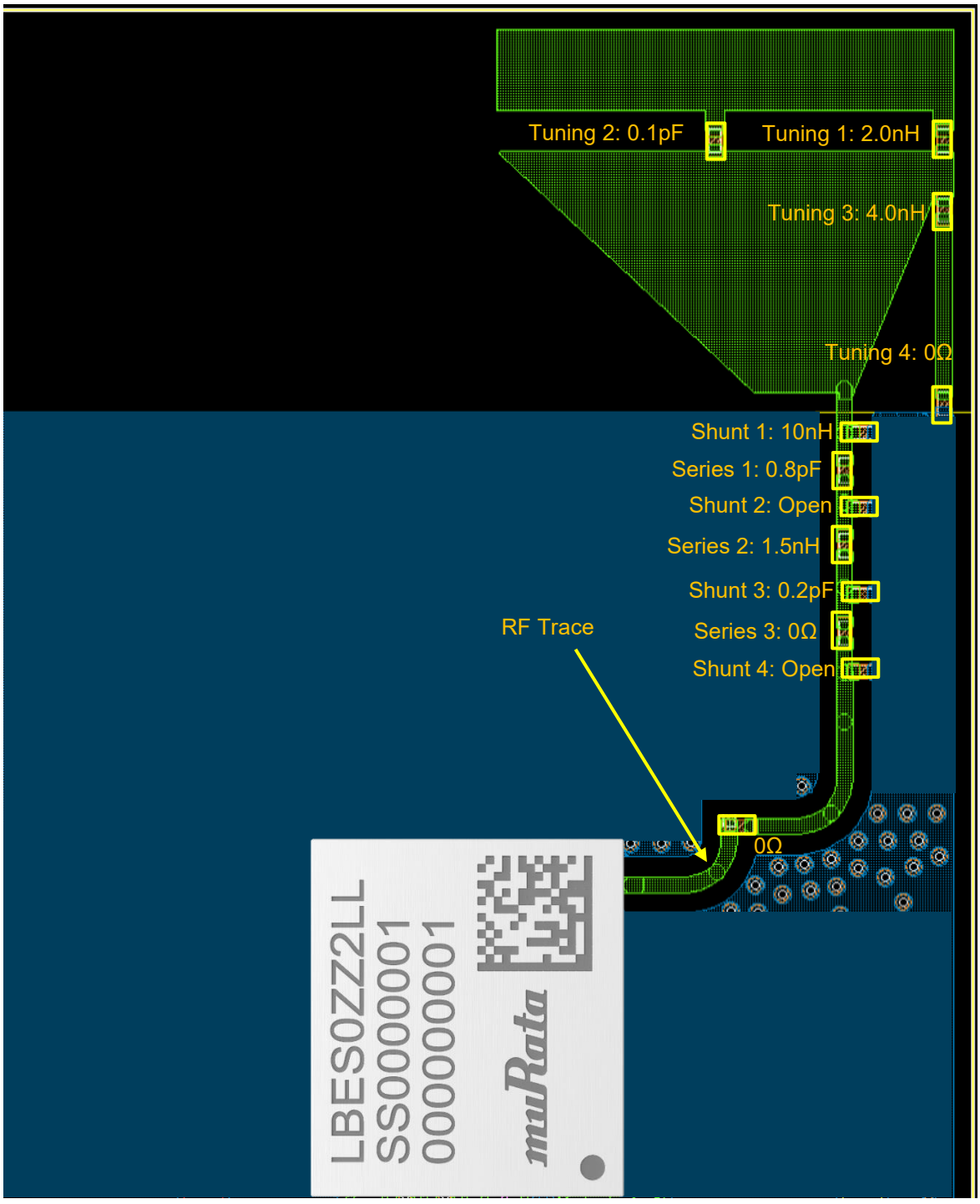


Table 6 shows reference SRF (Self-Resonant Frequency) values used during murata internal evaluations. Since each matching components is utilized for tuning within the specified frequency band, please select SRF values that are higher than the tuning band frequency.

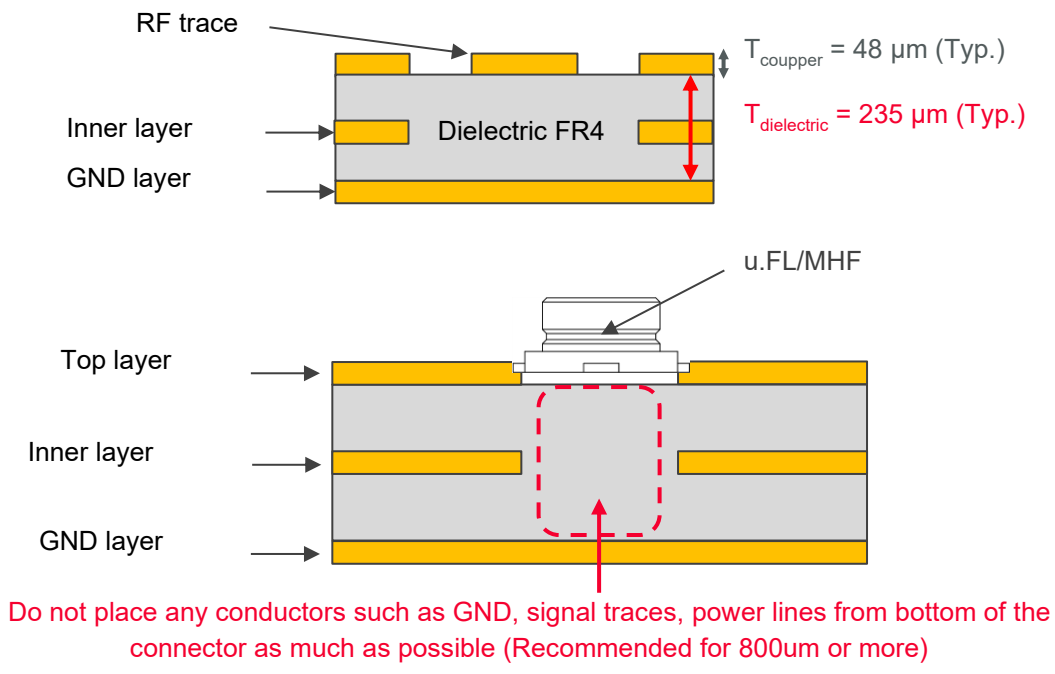
Table 6: Reference parameters of Trace Antenna tuning components

Reference No.	Tuning Band	Component Value	SRF
Tuning 1	2.4 GHz	2.0nH	12500MHz
Tuning 2	5 GHz	0.1pF	–
Tuning 3	2.4 GHz	4.0nH	5300MHz
Tuning 4	2.4 GHz	0ohm	–
Shunt 1	2.4 GHz	10nH	3200MHz
Series 1	2.4 GHz	0.8pF	–
Shunt 2	-	None	–
Series 2	5 GHz	1.5nH	15000MHz
Shunt 3	5 GHz	0.2pF	–
Series 3	2.4 G/5 GHz	0ohm	–
Shunt 4	-	None	–

3.6.3 PCB Stack-Up

Figure 7 shows the PCB stack-up layers.

Figure 7: PCB Stack-Up Layers



3.6.4 Trace Antenna Performance

This section illustrates the trace antenna performance results as shown in **Figure 8**.

Figure 8: Trace Antenna Performance

Gain and Efficiency

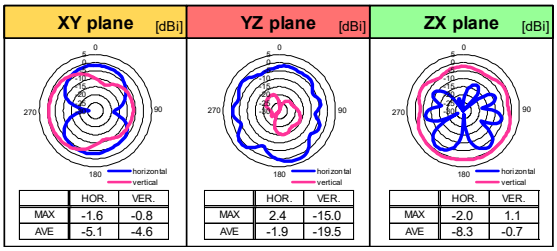
[dBi] [dB]

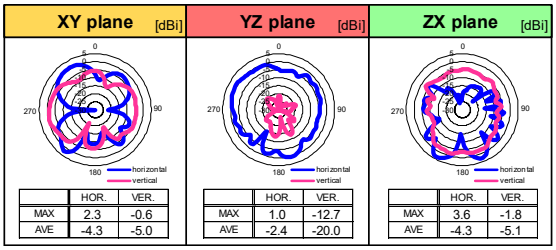
LINEAR POLARIZATION		XY-plane hor.	XY-plane ver.	YZ-plane hor.	YZ-plane ver.	ZX-plane hor.	ZX-plane ver.	Total Efficiency
2400 MHz	MAX.	-1.6	-0.9	2.6	-16.3	-2.2	1.0	
	AVE.	-4.9	-4.6	-2.0	-20.4	-8.3	-0.9	-1.0
2442 MHz	**MAX.**	**-1.6**	**-0.8**	**2.4**	**-15.0**	**-2.0**	**1.1**	
	AVE.	**-5.1**	**-4.6**	**-1.9**	**-19.5**	**-8.3**	**-0.7**	**-1.0**
2484 MHz	MAX.	-1.7	-0.7	2.5	-13.6	-1.7	1.6	
	AVE.	-5.2	-4.5	-1.6	-18.7	-8.2	-0.5	-0.9

[dBi] [dB]

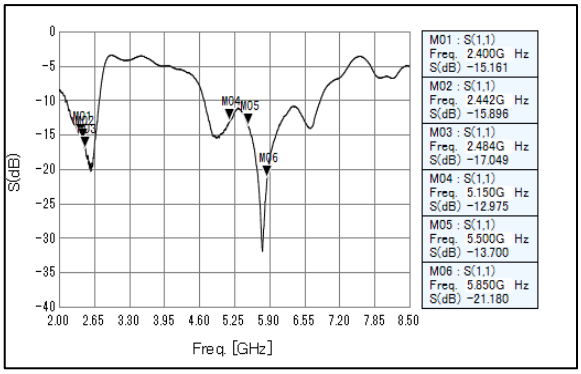
LINEAR POLARIZATION		XY-plane hor.	XY-plane ver.	YZ-plane hor.	YZ-plane ver.	ZX-plane hor.	ZX-plane ver.	Total Efficiency
5150 MHz	MAX.	2.3	0.1	2.2	-11.4	3.5	-0.2	
	AVE.	-4.1	-4.5	-2.0	-19.2	-3.9	-3.9	-1.3
5500 MHz	**MAX.**	**2.3**	**-0.6**	**1.0**	**-12.7**	**3.6**	**-1.8**	
	AVE.	**-4.3**	**-5.0**	**-2.4**	**-20.0**	**-4.3**	**-5.1**	**-1.6**
5850 MHz	MAX.	2.3	-0.7	1.0	-12.9	3.5	-1.6	
	AVE.	-4.1	-5.4	-2.4	-19.8	-4.2	-5.5	-1.5

Directivity





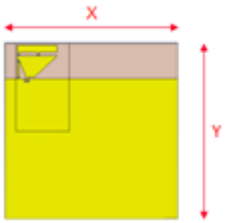
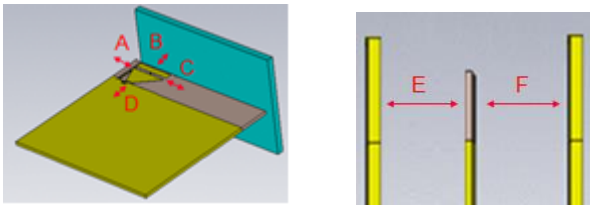
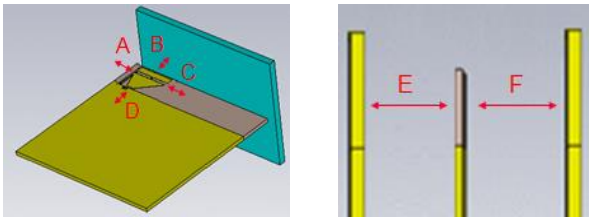
Return Loss



3.6.5 Trace Antenna Installation

Keep board size and clearance to Metal/GND and dielectric around the trace antenna for good antenna performance. **Table 7** lists the antenna installation details.

Table 7: Trace Antenna Installation

Board Size		X ≥ 40 mm Y ≥ 40 mm
Clearance to Metal/GND		A ≥ 20 mm B ≥ 20 mm C ≥ 20 mm D ≥ 20 mm E/F ≥ 20 mm
Clearance to Dielectric		A ≥ 4 mm B ≥ 4 mm C ≥ 4 mm D ≥ 4 mm E/F ≥ 4 mm

4 Setup Configuration Files

To enable Murata's regulatory certification, the configuration file below shall be loaded initially. The transmit power files are hosted at Murata GitHub for Linux.

4.1 WLAN Configuration Files for Linux

The files listed in **Table 8** shall be used to satisfy regulatory requirements if user wants to use Murata regulatory certification. For more regulatory information, refer to Linux User Guide.

Table 8: WLAN Configuration Files - Linux

Names	Country	Country Code	Configuration Files
WLAN Tx power configuration files	USA	US	txpower_US.bin
	Canada	CA	txpower_CA.bin
	Europe	DE	txpower_EU.bin
	Japan	JP	txpower_JP.bin
	Taiwan	TW	txpower_TW.bin
WLAN OFDMA RU Tx power configuration files	USA	US	rutxpower_US.bin
	Canada	CA	rutxpower_CA.bin
	Europe	DE	rutxpower_EU.bin
	Japan	JP	rutxpower_JP.bin
	Taiwan	TW	rutxpower_TW.bin
WLAN Carrier Sense / Adaptivity threshold configuration file			ed_mac.bin
WLAN regulatory limitation configuration file			db.txt

NXP IC based module shall use CRDA mechanism which is provided by Linux-wireless. Compile the new regulatory.bin file from “db.txt” which is provided by Murata as explained in wireless-regdb.

4.2 Bluetooth Configuration Files for Linux

Bluetooth Tx power configuration script files shall be loaded after Bluetooth device initialization.

The files listed in **Table 9** shall be used to satisfy regulatory requirements if user wants to use Murata regulatory certification.

Table 9: Bluetooth Configuration Files - Linux

Names	Country	Country Code	Script Files
Bluetooth Tx power configuration script files	USA	US	bt_power_config_US_CA_TW.sh
	Canada	CA	bt_power_config_US_CA_TW.sh
	Europe	DE	bt_power_config_EU.sh
	Japan	JP	bt_power_config_JP.sh
	Taiwan	TW	bt_power_config_US_CA_TW.sh

- Bluetooth Tx power configuration files
  - bt_power_config_US_CA_TW.sh
    The default OTP settings are configured for US/CA/TW transmission power setting; therefore, no script is required in these regions.
    However, if you need to dynamically change the settings to US/CA/TW from another location, please use the following command.
    - Command examples are as below:

```
# sh bt_power_config_US_CA_TW.sh
```

    - Content of shell script file “bt_power_config_US_CA_TW.sh”

```
hcitool -i hci0 cmd 0x03 0x003
hcitool -i hci0 cmd 0x3f 0x61 0x00 0x01 0x02 0x1C 0x37 0xEB 0x1C 0x00
0x1C 0x00 0x00 0x00 0x01 0x94 0x0A 0x04 0x28 0x00 0x00 0x00 0xC0 0xC6
0x2D 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x0A 0x64 0x8B 0x10 0x00
0xFF 0xFF 0xFF 0xFF 0x01 0x00 0x02 0x00 0x01 0x00 0x02 0x00
hcitool -i hci0 cmd 0x03 0x003
```

- bt_power_config_EU.sh
  - Command examples are as below:

```
# sh bt_power_config_EU.sh
```

  - Content of shell script file “bt_power_config_EU.sh”

```
hcitool -i hci0 cmd 0x03 0x003
hcitool -i hci0 cmd 0x3f 0x61 0x00 0x01 0x02 0x1C  0x37 0xEB 0x1C 0x00
0x1C 0x00 0x00 0x00 0x01 0x94 0x03 0x04 0x28 0x00 0x00 0x00 0xC0 0xC6
0x2D 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x0A 0x64 0x8B 0x10 0x00
0xFF 0xFF 0xFF 0xFF 0x01 0x00 0x02 0x00 0x01 0x00 0x02 0x00
hcitool -i hci0 cmd 0x03 0x003
```

- bt_power_config_JP.sh
  - Command examples are as below:

```
# sh bt_power_config_JP.sh
```

  - Content of shell script file “bt_power_config_JP.sh”

```
hcitool -i hci0 cmd 0x03 0x003
hcitool -i hci0 cmd 0x3f 0x61 0x00 0x01 0x02 0x1C 0x37 0xEB 0x1C 0x00
0x1C 0x00 0x00 0x00 0x01 0x94 0x05 0x04 0x28 0x00 0x00 0x00 0xC0 0xC6
0x2D 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x0A 0x64 0x8B 0x10 0x00
0xFF 0xFF 0xFF 0xFF 0x01 0x00 0x02 0x00 0x01 0x00 0x02 0x00
hcitool -i hci0 cmd 0x03 0x003
```

This script is based on baud rate 3000000.
Please be sure to run bt_power_config_**.sh script after the baud rate set as 3000000.

Bluetooth Tx power is configured by using hcitool.

4.3 802.15.4 Power Setting Command for Linux (2LL only)

802.15.4 Tx power setting command shall be implemented after 802.15.4 device initialization.

Tx power setting command shall be used to satisfy regulatory requirements if user wants to use Murata regulatory certification.

- US/CA/TW
  - Command examples are as below:

```
Ot-ctl txpower 10
```

- EU
  - Command examples are as below:

```
Ot-ctl txpower 3
```

- JP

- Command examples are as below:

```
Ot-ctl tx_power 8
```

4.4 Configuration Files for FreeRTOS

Please ask Murata at Murata Community Forum.

5 Reference Performance Data

5.1 Typical Rx Minimum Sensitivity Level at Module Antenna Port

This section describes the typical Rx minimum sensitivity level (at module antenna port) for WLAN and Bluetooth.

5.1.1 WLAN

- **Conditions:**
  - AVDD33 = 3.3V, SD_VIO = 1.8V, VIO = 3.3V
  - Combo FW: 18.80.5.p43

Table 10 and **Table 11** describe the typical Rx minimum sensitivity level at module antenna ports for WLAN at 2.4 GHz and 5 GHz.

Table 10: Rx Minimum Sensitivity Level - WLAN at 2.4 GHz (20 MHz)

Frequency in MHz	Rx Minimum Sensitivity Level [dBm]									
	11b		11g		11n (HT 20)		11ac (VHT 20)		11ax (HE 20)	
	1 Mbps	11 Mbps	6 Mbps	54 Mbps	MCS0	MCS7	MCS0	MCS8	MCS0	MCS9
2412	-98	-89	-93	-76	-92	-74	-92	-70	-92	-67
2442	-97	-89	-92	-75	-91	-73	-91	-69	-91	-67
2472	-98	-89	-92	-75	-91	-72	-91	-69	-91	-66

Table 11: Rx Minimum Sensitivity Level - WLAN at 5 GHz (20 MHz)

Frequency in MHz	Rx Minimum Sensitivity Level [dBm]							
	11a		11n (HT 20)		11ac (VHT 20)		11ax (HE 20)	
	6 Mbps	54 Mbps	MCS0	MCS7	MCS0	MCS8	MCS0	MCS9
5180	-92	-75	-91	-73	-91	-69	-91	-66
5500	-91	-75	-91	-73	-91	-69	-91	-66
5825	-91	-74	-91	-73	-91	-69	-91	-66

5.1.2 Bluetooth

- **Conditions:**
  - AVDD33 = 3.3V, SD_VIO = 1.8V, VIO = 3.3V
  - Combo FW: 18.80.5.p43

Table 12 describes the typical Rx minimum sensitivity level for Bluetooth.

Table 12: Rx Minimum Sensitivity Level - Bluetooth

Frequency in MHz	Rx Minimum Sensitivity level[dBm]			
	LE 1M	LE 2M	125K	500K
2412	-103	-100	-107	-108
2442	-103	-100	-106	-108
2472	-103	-100	-106	-108

5.1.3 802.15.4(2LL only)

- **Conditions:**
  - AVDD33 = 3.3V, SD_VIO = 1.8V, VIO = 3.3V
  - Combo FW: 18.80.5.p43

Table 13 describes the typical Rx minimum sensitivity level for 802.15.4.

Table 13: Rx Minimum Sensitivity Level - 802.15.4

Frequency in MHz	Rx Minimum Sensitivity level[dBm]
	802.15.4
2412	-104
2442	-104
2472	-103

5.2 Typical Tx/Rx Current Consumption

This section describes the typical Tx/Rx current consumption for WLAN and Bluetooth.

5.2.1 WLAN

Typical Tx/Rx current consumption for Wi-Fi has certain conditions as described below.

- **Conditions:**
  - AVDD33 = 3.3V, SD_VIO = 1.8V, VIO = 3.3V
  - Combo FW: 18.80.5.p43

Figure 9 shows a typical graph of Tx/Rx current consumption for Wi-Fi.

Figure 9: Typical Tx/Rx Current Consumption - Wi-Fi

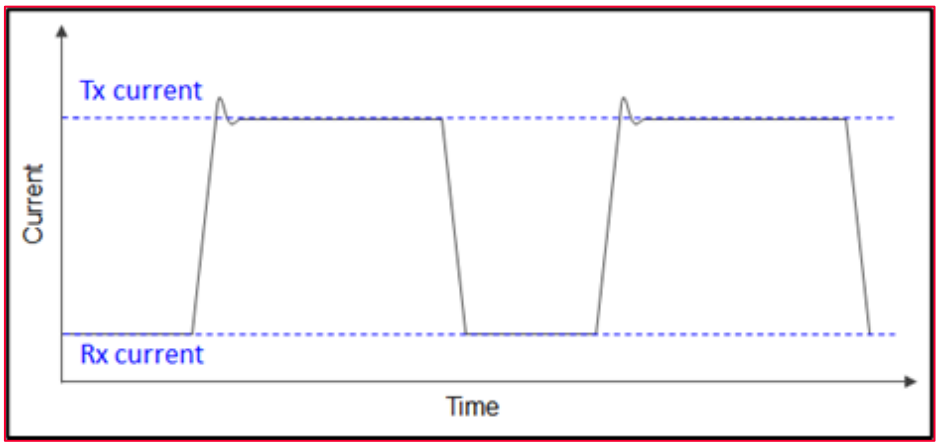


Table 14 describes the typical Tx/Rx current consumption for WLAN at 2.4 GHz.

Table 14: Typical Tx/Rx Current Consumption - WLAN at 2.4 GHz

Mode	Data Rate	Setting Tx Power [dBm]	Current [mA]	
			Tx	Rx
			AVDD33	AVDD33
11b	1 Mbps	17	260	59
11g	6 Mbps	16	250	59
11n (HT20)	MCS0	15	240	59
11ac (VHT20)	MCS0	15	240	59
11ax (HE20)	MCS0	15	240	59

Table 15 describes the typical Tx/Rx current consumption for WLAN at 5 GHz.

Table 15: Typical Tx/Rx Current Consumption - WLAN at 5 GHz

Mode	Data Rate	Setting Tx Power [dBm]	Current [mA]	
			Tx	Rx
			AVDD33	AVDD33
11a	6 Mbps	16	410	60
11n (HT20)	MCS0	15	390	60
11ac (VHT20)	MCS0	15	390	60
11ax (HE20)	MCS0	15	390	60

5.2.2 Bluetooth

Typical Tx/Rx current consumption for Bluetooth has certain conditions as described below.

- **Conditions:**
  - AVDD33 = 3.3V, SD_VIO = 1.8V, VIO = 3.3V
  - Combo FW: 18.80.5.p43

Figure 10 shows the typical graph of Tx/Rx current consumption for Bluetooth.

Figure 10: Typical Tx/Rx Current Consumption - Bluetooth

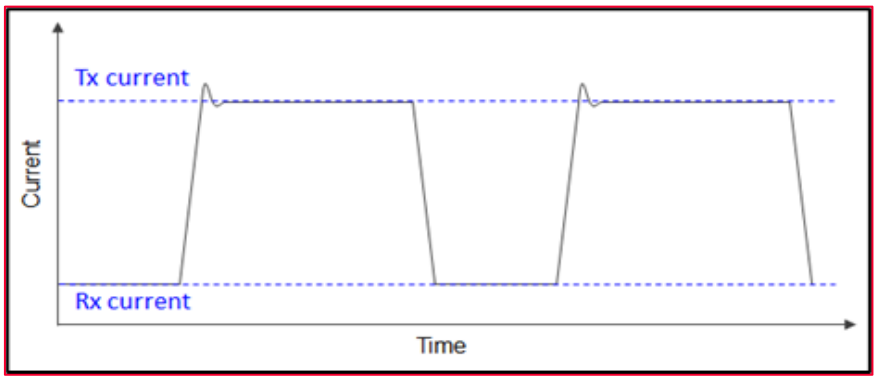


Table 16 describes the typical Tx/Rx current consumption for Bluetooth.

Table 16: Typical Tx/Rx Current Consumption - Bluetooth

Mode	Setting Tx Power [dBm]	Current [mA]	
		Tx	Rx
		AVDD33	AVDD33
LE 1M	10	50	18
LE 2M	10	50	18

5.2.3 802.15.4 (2LL only)

Typical Tx/Rx current consumption for Bluetooth has certain conditions as described below.

- **Conditions:**
  - AVDD33 = 3.3V, SD_VIO = 1.8V, VIO = 3.3V
  - Combo FW: 18.80.5.p43

Figure 11 shows the typical Tx/Rx current consumption for 802.15.4.

Figure 11: Typical Tx/Rx Current Consumption - 802.15.4

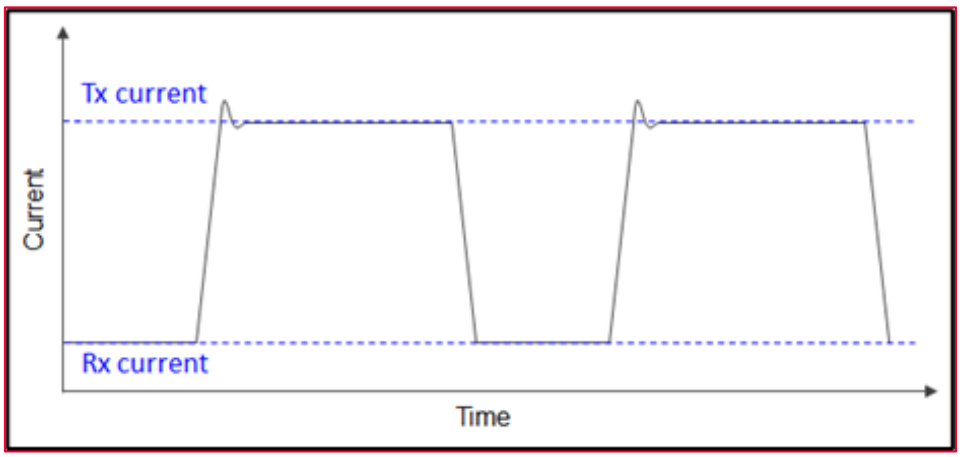


Table 17 describes the typical Tx/Rx current consumption for 802.15.4.

Table 17: Typical Tx/Rx Current Consumption - 802.15.4

Mode	Setting Tx Power [dBm]	Current [mA]	
		Tx	Rx
		AVDD33	AVDD33
802.15.4	10	55	18

5.3 Typical Sleep Current Consumption

This section describes the typical sleep current consumption for Wi-Fi and Bluetooth.

5.3.1 WLAN

- **Conditions:**
  - AVDD33 = 3.3V, SD_VIO/VIO = 1.8V
  - Target: NXP 8MMINILPD4-EVKB
  - WLAN FW: 18.99.5.p43
  - Beacon Interval = 100 ms
  - SDIO Signal Level = 1.8V
  - Frame duration for short beacon IEEE-PS current measurement:
    2.4 GHz: 1000 us
    5 GHz: 300us
  - Measurement Equipment for inter-beacon: DC Power Analyzer N6705C

Table 18 describes the typical sleep current consumption for WLAN.

Table 18: Typical Sleep Current Consumption - WLAN

Current Consumption AVDD33 (3.3V) in mA			
Band	**SD clock speed**	**SDIO 2.0**	**SDIO 3.0**
	Chip deep sleep	0.3	0.97
2.4 GHz	IEEE Power Save: DTIM1	0.86	1.48
	IEEE Power Save: DTIM3	0.49	1.14
	IEEE Power Save: DTIM5	0.41	1.07
5 GHz	IEEE Power Save: DTIM1	0.67	1.31
	IEEE Power Save: DTIM3	0.42	1.09
	IEEE Power Save: DTIM5	0.37	1.04

5.3.2 Bluetooth

- **Conditions:**
  - AVDD33 = 3.3V, VIO = 1.8V
  - Target: NXP 8MMINILPD4-EVKB
  - Bluetooth/15.4 FW: 18.99.5.p43
  - Bluetooth Interface: UART

Table 19 describes the typical sleep current consumption for Bluetooth.

Table 19: Typical Sleep Current Consumption - Bluetooth

Current Consumption AVDD33 (3.3V) in mA	
Bluetooth only in deep sleep mode	0.2

5.4 Typical Bluetooth Current Consumption

This section describes the typical Bluetooth current consumption.

- **Conditions**:
  - AVDD33 = 3.3V, SD_VIO/VIO = 1.8V
  - Target: NXP 8MMINILPD4-EVKB
  - Bluetooth/15.4 FW: 18.99.5.p43
  - Bluetooth Interface: UART

Table 20 describes the typical Bluetooth advertise current consumption.

Table 20: Typical Bluetooth Current Consumption

Current Consumption AVDD18 (1.8V) in mA	
Bluetooth LE advertise / Interval = 1.28sec	0.27
Bluetooth LE scan / interval = 1.28sec	0.34

5.5 Typical Throughput

The typical throughput test configurations are:

- AVDD18 = 1.8V, AVDD33 = 3.3V, SD_VIO/VIO = 1.8V
- Target: NXP 8MMINILPD4-EVKB
- Combo FW: 18.99.5.p43
- Access Point: RT-AX88U (ASUS)
- Distance between Access Point and the Target is around 3 ft.
- UDP commands**:** Bit rate was set at more than 20% of observed corresponding TCP throughput.
  Sample UDP command.

```
iperf3 <server-ip-addr> -u -b <20%-of-TCP>M -P1 -t 60
```

Table 21 describes the typical throughput data for the modules.

Table 21: WLAN Typical Throughput Data

Mode	TCP Throughput in Mbps		UDP Throughput in Mbps	
	Tx	Rx	Tx	Rx
2.4 GHz 11ax HE20	89	96	97	100
5 GHz 11ax HE20	95	99	101	104

6 References

Table 22 reviews all the key reference documents that the user may like to refer to.

Table 22: Reference Table

Support Site	Notes
Murata Type 2LL Module Datasheet	Murata Type 2LL module datasheet (TYPE2LL.pdf)
Murata Type 2KL Module Datasheet	Murata Type 2KL module datasheet (TYPE2KL.pdf)
Murata Type 2LL 2KL Module Footprint	Murata Type 2LL / 2KL module footprint (type2kl-2ll-module-footprint-topview.dxf)
Murata Type 2LL 2KL PCB Type Di-pole and Trace Antenna	Murata Type 2LL / 2KL module trace antenna (type2kl-2ll-PCB-Type-Di-pole-and-Trace-Antenna.dxf)
Linux	Murata GitHub link for Linux transmit power files for 2LL /2KL
wireless-regdb	Regulatory database used by Linux
Linux User Guide	Murata Linux User Guide for NXP modules (Murata Wi-Fi & BT (NXP) Solution for i.MX Linux User Guide.pdf). Murata website to be updated soon.



In case Murata website does not have the updated document, please refer to the Connectivity Module page on the Murata Community Forum. This contains a pinned post with all the updated documents.

7 Technical Support Contacts

Table 23 lists all the support resources available for the Murata Wi-Fi/BT solution.

Table 23: List of Support Resources

Support Site	Notes
Murata Community Forum	**Primary support point for technical queries.** This is an open forum for all customers. Registration is required.
Murata i.MX Landing Page	No login credentials required. Murata documentation covering hardware, software, testing, etc. is provided here.
Murata uSD-M.2 Adapter Landing Page	Landing page for uSD-M.2 Adapter. In conjunction with Murata i.MX Landing Page, this should provide the user with comprehensive getting started documentation.
Murata Module Landing Page	No login credentials required. Murata documentation covering all Infineon-based Wi-Fi/BT modules is provided here.

Revision History

Revision	Date	Changed Item	Comment
1.0	Apr 07, 2025	First Issue	
2.0	Aug 21, 2025	3.1 Reference Circuit 3.6.2 Trace Antenna 4.2 Bluetooth Configuration Files for Linux	• Type 2KL Pin 6 from PCM_SYNC to GPIO[7] • Added reference SRF value • Added US/CA/TW script file
3.0	May 22, 2026	4.2 Bluetooth Configuration Files for Linux	• Update Script. Support 3M baud rate.



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